Module LSP_FALL_2024
Package Org.howard.edu.lsp.assignment5

Class IntegerSet

java.lang.Object
Org.howard.edu.lsp.assignment5.IntegerSet

```
public class IntegerSet
extends Object
```

Constructor Summary

Constructors

Constructor	Description
`IntegerSet()`	
`IntegerSet(ArrayList <Integer > set)`	

Method Summary

All Methods Instance Methods Concrete Methods

Modifier and Type	Method	Description
void	`add(int item)`	Adds an item to the set or does nothing if it is already there.
void	`clear()`	Clears the internal representation of the set.
void	`complement(IntegerSet intSetb)`	Performs a set complement operation with another IntegerSet.
boolean	`contains(int value)`	Returns true if the set contains the value, otherwise false.
void	`diff(IntegerSet intSetb)`	Performs a set difference operation with another IntegerSet.
boolean	`equals(Object  o)`	Returns true if the 2 sets are equal, false otherwise.
void	`intersect(IntegerSet intSetb)`	Performs a set intersection operation with another IntegerSet.

boolean	isEmpty()	Returns true if the set is empty, false otherwise.
int	largest()	Returns the largest item in the set.
int	length()	Returns the length of the set (number of elements).
void	remove(int item)	Removes an item from the set or does nothing if it is not there.
int	smallest()	Returns the smallest item in the set.
String	toString()	Returns a String representation of the set.
void	union(IntegerSet intSetb)	Performs a set union operation with another IntegerSet.

Methods inherited from class java.lang.Object

getClass , hashCode , notify , notifyAll , wait , wait , wait

Constructor Details

IntegerSet

```
public IntegerSet()
```

IntegerSet

```
public IntegerSet(ArrayList <Integer > set)
```

Method Details

clear

```
public void clear()
```

Clears the internal representation of the set.

length

```
public int length()
```

Returns the length of the set (number of elements).

Returns:

The length of the set.

equals

```
public boolean equals(Object  o)
```

Returns true if the 2 sets are equal, false otherwise. Two sets are equal if they contain all of the same values in ANY order. This overrides the equals method from the Object class.

Overrides:

`equals` in class `Object`

Parameters:

`o` - The object to compare to.

Returns:

True if the sets are equal, false otherwise.

contains

```
public boolean contains(int value)
```

Returns true if the set contains the value, otherwise false.

Parameters:

`value` - The value to check for.

Returns:

True if the set contains the value, false otherwise.

largest

```
public int largest()
```

Returns the largest item in the set.

Returns:

The largest item in the set.

Throws:

`IllegalStateException` - If the set is empty.

smallest

```
public int smallest()
```

Returns the smallest item in the set.

Returns:

The smallest item in the set.

Throws:

`IllegalStateException` - If the set is empty.

add

```
public void add(int item)
```

Adds an item to the set or does nothing if it is already there.

Parameters:

`item` - The item to add.

remove

```
public void remove(int item)
```

Removes an item from the set or does nothing if it is not there.

Parameters:

`item` - The item to remove.

union

```
public void union(IntegerSet intSetb)
```

Performs a set union operation with another IntegerSet. This set will contain all elements from both sets.

Parameters:

`intSetb` - The other IntegerSet to perform the union with.

intersect

```
public void intersect(IntegerSet intSetb)
```

Performs a set intersection operation with another IntegerSet. This set will contain only the elements that are common to both sets.

Parameters:

`intSetb` - The other IntegerSet to perform the intersection with.

diff

```
public void diff(IntegerSet intSetb)
```

Performs a set difference operation with another IntegerSet. This set will contain only the elements that are in this set but not in the other set.

Parameters:

`intSetb` - The other IntegerSet to perform the difference with.

complement

```
public void complement(IntegerSet intSetb)
```

Performs a set complement operation with another IntegerSet. This set will contain all the elements that are in the other set but not in this set.

Parameters:

`intSetb` - The other IntegerSet to perform the complement with.

isEmpty

```
public boolean isEmpty()
```

Returns true if the set is empty, false otherwise.

Returns:

True if the set is empty, false otherwise.

toString

```
public String  toString()
```

Returns a String representation of the set.

Overrides:

`toString` in class `Object`

Returns:

A String representation of the set.